



Tyres that pump themselves up

An under-inflated tyre can reduce mileage, increase wear-and-tear and even burst. However, during long journeys, when tyres are likely to wear out quickly, drivers are in no mood to monitor their tyre pressure. As a brilliant alternative, Goodyear has come up with a self-regulating tyre that inflates itself on-the-go. The electronics and mechanical components are built into the tyre itself, and the technology is currently being used on 100-psi, re-treaded commercial usage tyres.

Inside: Goodyear's Air Maintenance Technology focuses on an internal regulator comprising pressure sensors and miniature pump, completely fitted inside the tyre itself. The power for regulator functioning is drawn from the movement of the tyres. Working is simple: A sensor on the interior detects when the pressure goes low. Immediately a valve mounted on the sidewall opens up. As the tyre rotates, the vehicle's weight compresses a tube built into the tyre's circumference to inject air into the tyre. Once the pressure reaches the programmed psi value, the system shuts the valve and the tyre runs as usual.

A social hearing-aid that imitates functional hearing

People using hearing-aids are often embarrassed by the lack of clarity in crowded places. Two researchers at the University of Toronto have floated a start-up called Cogni-Wave to achieve what they call 'social hearing.' Cogni-Wave is developing a hearing-aid that uses rotating microphones to filter environmental sounds, so that the person using the hearing-aid hears exactly like someone with functional hearing—one voice at a time.

"We have a prototype that utilises Bluetooth and radio frequency (RF) wireless microphones or headphones. We have had good results but we are still adjusting parameters and the system to optimise performance. We are implementing our algorithm onto hardware, that is, chips and circuits, to speed up the processing and better incorporate it into various forms of hearing enhancement devices," says Brian Wang, co-founder, Cogni-Wave. These hearing-aids will hopefully hit the markets by the end of 2013.

Inside: Michael Shen, co-founder, Cogni-Wave, says, "We will use digital signal processor (DSP) chips for the core signal processing of our device; the specifics of the chips would match our algorithm parameters to achieve desired results. Directional microphone and miniaturised speakers are necessary for our design, along with the circuitry components needed to amplify or filter the received and outgoing signals."



A camera that takes 'living' pictures

Lytro is the first consumer camera that captures the entire light field – that is, the rays of light travelling in every direction through a scene. Earlier, this used to be done in labs with hundreds of cameras and a supercomputer to collate the images. The scientists who made Lytro have combined all that technology into one small handy camera! This means you can take 'living' pictures that you can interactively refocus later, any number of times. There is a special format and software bundle that let you do a lot of things with your living pictures, and also share them with others.

Inside: Lytro's 8X optical-zoom, f/2 lens captures significantly more light than most cameras. A micro-lens array joined to a digital image sensor records the colour, intensity and direction of the 11 million or so light rays entering the camera. Their light field engine does the work of a super-computer, by processing all the light ray information and creating images that can be refocused anywhere, anytime. Since the engine travels with every file, others can also refocus shared images. The camera comes in a lightweight anodised aluminium structure with a touch screen.